# ***Analysis Done in AI & LLM Jobs***

1) How many Jobs are there in each Sector?

```
select sector, count(*) as No_of_Jobs from jobs group by sector;
```

2) No. of jobs as per jobs_title?

```
select job_title, count(*) as No_of_Jobs from jobs group by job_title;
```

3) Top 5 jobs as per jobs_title? / Most Common Job Title

```
select job_title, count(*) as No_of_Jobs from jobs
group by job_title order by No_of_Jobs desc limit 5 ;
```

4) What is the average salary?

```
select job_title,round(avg(salary_usd_annual),2) as Average_Salary
from jobs group by job_title order by Average_Salary desc;
```

5) What is the average experience requirement?

```
select job_title,round(avg(years_experience_min),2) as Average_Experience
from jobs group by job_title order by Average_Experience asc;
```

6) How many jobs are remote/hybrid/on-site?

```
select remote_type,count(*)as No_of_Jobs from jobs
group by remote_type;
```

7) Average salary

```
select round(avg(salary_usd_annual),2) from jobs;
```

8) Median salary

```
with ranked as(
select salary_usd_annual, row_number() over(order by salary_usd_annual) as sort, count(*) over()
as total from jobs)
select round(avg(salary_usd_annual),2)as median_salary from ranked where sort
in(floor((total+1)/2),ceil((total+1)/2));
```

9) Highest-paying jobs

```
select job_id,job_title,company,sector,salary_usd_annual from jobs
where salary_usd_annual =(select max(salary_usd_annual)as highest_salary from jobs);
```

or

```
select job_id,job_title,company,sector,salary_usd_annual from jobs
order by salary_usd_annual desc limit 1;
```

## 10) Salary by country

```
select country, avg(salary_usd_annual)as average_salary_usd,
max(salary_usd_annual)as highest_salary_usd,
min(salary_usd_annual) as min_salary_usd,
count(salary_usd_annual)as total_vacancy from jobs group by country order by country;
```

## 11) Salary by sector

```
select sector, avg(salary_usd_annual) Avg_Salary_USD,
max(salary_usd_annual) as Highest_Salary_USD,
min(salary_usd_annual) as Lowest_Salary_USD,
count(*) as No_of_Jobs from jobs group by sector order by sector;
```

## 12) Salary by experience

```
select experience_level,round(avg(salary_usd_annual),2) as Avg_Salary_USD,
max(salary_usd_annual) as Higest_Salary_USD,
min(salary_usd_annual) as Lowest_Salary_USD,
count(*) as Total_Jobs from jobs group by experience_level order by experience_level;
```

## 13) Salary by work mode

```
select remote_type,round(avg(salary_usd_annual),2) as Avg_Salary_USD,
max(salary_usd_annual) as Higest_Salary_USD,
min(salary_usd_annual) as Lowest_Salary_USD,
count(*) as Total_Jobs from jobs group by remote_type order by remote_type;
```

## 14) Salary by company size

```
select company, avg(salary_usd_annual) as Avg_Salary_USD,
max(salary_usd_annual) as Highest_Salary_USD,
min(salary_usd_annual) as Lowest_Salary_USD,
count(*) as No_of_Jobs from jobs  group by company order by company;
```

## 15) Contract vs full-time salary

```
select is_contract, round(avg(salary_usd_annual),2) as Avg_Salary,
max(salary_usd_annual) as Highest_Salary_USD,
min(salary_usd_annual) as Lowest_salary_USD,
count(*) as No_of_Jobs from jobs
where is_contract in ("Full-time","Contract")
group by is_contract ;
```

## 16) Salary differences for jobs requiring different skills

```
select 'DBT' as Skill, round(avg(salary_usd_annual),2) as Avg_Salary,max(salary_usd_annual) as
Highest_salary,
min(salary_usd_annual) as lowest_salary,count(*) as No_of_Jobs
from jobs where required_skills like '%dbt%'
```

```
union all
select 'Fast API',round(avg(salary_usd_annual),2) as Avg_Salary,max(salary_usd_annual) as
Highest_salary,
min(salary_usd_annual) as lowest_salary,count(*) as No_of_Jobs
from jobs where required_skills like '%fastapi%'
union all
select 'Azure ML',round(avg(salary_usd_annual),2) as Avg_Salary,max(salary_usd_annual) as
Highest_salary,
min(salary_usd_annual) as lowest_salary,count(*) as No_of_Jobs
from jobs where required_skills like '%Azure ml%'
union all
select 'CUDA',round(avg(salary_usd_annual),2) as Avg_Salary,max(salary_usd_annual) as
Highest_salary,
min(salary_usd_annual) as lowest_salary,count(*) as No_of_Jobs
from jobs where required_skills like '%cuda%'
union all
select 'Hugging Face',round(avg(salary_usd_annual),2) as Avg_Salary,max(salary_usd_annual) as
Highest_salary,
min(salary_usd_annual) as lowest_salary,count(*) as No_of_Jobs
from jobs where required_skills like '%hugging face%'
union all
select 'Kubernetes',round(avg(salary_usd_annual),2) as Avg_Salary,max(salary_usd_annual) as
Highest_salary,
min(salary_usd_annual) as lowest_salary,count(*) as No_of_Jobs
from jobs where required_skills like '%kubernetes%'
union all
select 'Pytorch',round(avg(salary_usd_annual),2) as Avg_Salary,max(salary_usd_annual) as
Highest_salary,
min(salary_usd_annual) as lowest_salary,count(*) as No_of_Jobs
from jobs where required_skills like '%pytorch%'
union all
select 'JAX',round(avg(salary_usd_annual),2) as Avg_Salary,max(salary_usd_annual) as
Highest_salary,
min(salary_usd_annual) as lowest_salary,count(*) as No_of_Jobs
from jobs where required_skills like '%JAX%'
union all
select 'RAG',round(avg(salary_usd_annual),2) as Avg_Salary,max(salary_usd_annual) as
Highest_salary,
min(salary_usd_annual) as lowest_salary,count(*) as No_of_Jobs
from jobs where required_skills like '%rag%'
union all
select 'ML Flow',round(avg(salary_usd_annual),2) as Avg_Salary,max(salary_usd_annual) as
Highest_salary,
min(salary_usd_annual) as lowest_salary,count(*) as No_of_Jobs
from jobs where required_skills like '%MLflow%' order by Skill;
```

## 17) Required skills for Job Titles

```
select case when required_skills like'%OpenAI API%' or required_skills like'%LangChain%'
or required_skills like'%RAG%' or required_skills like'%Fine-tuning,%' or required_skills like'%rlhf%'
or required_skills like'% Hugging Face Transformers%' or required_skills like'%LlamaIndex%'
then 'Generative AI / LLM Engineering'

when required_skills like'%PyTorch%' or required_skills like'%TensorFlow,%' or required_skills
like'%JAX%'
or required_skills like'%CUDA%' then 'Machine Learning Engineering'

when required_skills like'%MLflow%' or required_skills like'%AWS SageMaker,%' or required_skills
like'%Azure ML%'
or required_skills like'%GCP Vertex AI%' or required_skills like'%Weights & Biases%' or
required_skills like'%Docker,%'
or required_skills like'%Kubernetes%' then 'MLOps / AI Platform Engineering'

when required_skills like'%Pinecone' or required_skills like'%Weaviate%' or required_skills
like'%ChromaDB,%' or
required_skills like'%Vector Databases%' then 'Vector Database / RAG Infrastructure'

when required_skills like'%Python%' or required_skills like'%FastAPI%' or required_skills
like'%Rust,%' or required_skills like'%c++%'
or required_skills like'%spark%' or required_skills like'%sql%' or required_skills like'%dbt,%'
then 'AI Backend / Infrastructure'

else 'Other' end as Skills ,
count(*) as No_of_Jobs, max(salary_usd_annual) as Highest_Salary_USD
,min(salary_usd_annual) as Lowest_Salary, round(avg(salary_usd_annual),2) as Avg_salary_usd
from jobs group by Skills order by  Skills;
```

## 18) Primary LLM Mentioned

```
select primary_llm_mentioned as LLM, count(*) as No_of_Company_Used_LLM
from jobs group by primary_llm_mentioned order by No_of_Company_Used_LLM;
```

## 19) How many Jobs for these Specific Skills

```
select Skills, count(*) as No_of_jobs from jobs cross join(
select 'Python' as Skills union all select 'RAG' union all select 'OpenAI API' union all select
'LangChain'
union all select 'Pytorch' union all select 'TensorFlow' union all select 'Hugging Face Transformers'
union all select 'Pinecone' union all select 'Mlfow' union all select 'AWS Sagemaker') as All_skills
where required_skills like concat('%',Skills,'%') group by skills order by No_of_jobs;
```

## 20) Most Required Skills

```
with recursive skill_split as(
select trim(substring_index(required_skills,',',1)) as skill,
case when required_skills like '%,%'
  then trim(substring(required_skills,locate(',',required_skills)+1))
else ''  end as remaining_skills from jobs

union all
select trim(substring_index(remaining_skills,',',1)) ,
case when remaining_skills like '%,%'
then trim(substring(remaining_skills,locate(',',remaining_skills)+1))
else '' end  from skill_split where remaining_skills<> ''
)
select skill,count(*) as skill_sount from skill_split group by 1 order by 2 desc limit 10;
```

## 21) Job Titles required LLM as per Company

```
select job_title, primary_llm_mentioned, count(*) No_of_Jobs,
round(avg(salary_usd_annual),2) as Avg_Salary_USD from jobs
group by job_title, primary_llm_mentioned order by job_title;
```

## 22) Experience vs Salary

```
select experience_years, round(avg(salary_usd_annual),2) as AVG_Salary_USD,
min(salary_usd_annual) Low_Salary, max(salary_usd_annual) as Highest_salary_USD,
count(*) as No_of_Jobs from jobs group by experience_years order by 1;
```

## 23) Company Offering Jobs as per Educations

```
select education_required as Education,
count(*) as No_of_Jobs,
count(distinct company) No_of_Companies
from jobs group by 1 order by 1 ;
```

## 24) Top 10 Companies hiring the most

```
 select company, count(*) No_of_Hiring
from jobs group by 1 order by 2 desc limit 10;
```

## 25) Companies Offering the Single Highest Salaries

```
select company, max(salary_usd_annual) as Higest_salary
from jobs group by 1 order by 2 desc;
```

## 26) Companies Offering the Average Highest Salaries

```
select company, round(avg(salary_usd_annual),2) as Avg_Salary
from jobs group by 1 order by 2 desc limit 10;
```

## 27) Company size vs Salary

```
select company_size, round(avg(salary_usd_annual),2) as Avg_Salary,
max(salary_usd_annual) as Highest_salary,min(salary_usd_annual) as Lowest_salary,
count(*) No_of_Jobs
from jobs group by 1 order by 2 desc limit 10;
```

## 28) Company size vs equity

```
select company_size,
count(*) No_of_Jobs,sum(case when has_equity ='Yes' then 1 else 0 end) as With_Equity,
round(100 * sum(case when has_equity ='Yes' then 1 else 0 end)/count(*),2) as Equity_Percentage
from jobs group by company_size order by company_size asc;
```

## 29) Company-Size VS Visa Sponsorship

```
select company_size, count(*) as Total_Jobs,
sum(case when visa_sponsorship = 'yes' then 1 else 0 end) as With_Visa_Sponsorship,
sum(case when visa_sponsorship='no' then 1 else 0 end) as Without_Visa_Sponsorship,
round(100*sum(case when visa_sponsorship='yes' then 1 else 0 end)/count(*),2)as
Percentage_of_Getting_Sponsorship
from jobs group by 1 order by 1;
```

## 30) Company VS Visa Sponsorship

```
select company, count(*) as Total_Jobs,
sum(case when visa_sponsorship = 'yes' then 1 else 0 end) as With_Visa_Sponsorship,
sum(case when visa_sponsorship='no' then 1 else 0 end) as Without_Visa_Sponsorship,
round(100*sum(case when visa_sponsorship='yes' then 1 else 0 end)/count(*),2)as
Percentage_of_Getting_Sponsorship
from jobs group by 1 order by 1;
```

## 31) Contract vs non-contract jobs

```
select case when is_contract='contract' then 'Contract'
when is_contract in ('Full-time','part-time') then 'Non_contract' end as Job_Types,
round(avg(salary_usd_annual),2) as Avg_Salary_USD, min(salary_usd_annual) as Low_Salary_USD,
max(salary_usd_annual) as High_salary_USD, count(*) as Total_Jobs from jobs
group by 1 order by 5 desc;
```

## 32) Equity availability

```
select has_equity as Equity_status,count(*) as Total_jobs,
round(count(*)*100/(select count(*) from jobs),2) as Job_percentage,
round(avg(salary_usd_annual),2) as AVG_Salary_USD from jobs group by 1;
```

## 33) Visa sponsorship availability

```
select visa_sponsorship as Visa_ Sponsorship_status,count(*) as Total_jobs,
round(count(*)*100/(select count(*) from jobs),2) as Job_percentage,
round(avg(salary_usd_annual),2) as AVG_Salary_USD from jobs group by 1;
```